

Simulation of Limit Order Book Dynamics with a Matching Engine

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Section 1. Literature Review

In modern high-frequency markets, limit order books are essential for creating fair, transparent and efficient financial markets. A limit order book (LOB) is a real-time electronic data that records all unexecuted buy and sell orders. The matching engine is the core algorithm that continuously matches the buy and sell orders in the LOB to ensure price-time priority (First-In, First-Out).

Early studies on financial markets dynamics adopted agent-based models to explain complex market facts by imitating heterogeneous investors. Classic examples like Lux and Marchesi (1999) support the idea that scaling and volatility clustering as a result of agent interactions. Preis et al. (2006) first built a multi-agent model with an explicit limit order book to study how simple trading rules can create complex market behavior. In their model, agents place markets and limit orders based on probabilistic rules. These interactions lead to the behavior of Hurst exponent and "fat tails" return distributions, just like the patterns we see in real markets.

Following the agent-based tradition, researchers tried to simulate limit order book more analytically and focused on reproducing the statistical properties. These models typically describe the order book as a queueing system, where limit orders arrive according to Poisson processes and are removed either by execution or cancellation. One of the foundational works in this area is Cont, Stoikov, and Talreja (2010), who proposed a continuous-time stochastic model for order book dynamics. They use parameter estimation and Laplace transformation to calculate the conditional probability of order book state, capturing the evolution of bid-ask spreads, order imbalances, and mid-price movements. Muni Toke (2015) developed a limit order book model based on simple queueing theory, where order arrivals follow a Poisson process. Using a spatial Poisson framework, he built a continuous version of the model and derived an analytical formula for the average shape of the order book. The model also introduces randomness in order sizes, assuming they follow a geometric distribution, and shows that the relative size of limit orders has a significant effect on the book's average depth. Using data from stocks on the Paris Stock Exchange, he finds empirical evidence that the cumulative depth of the order book is positively correlated with overall market activity.

After the early Poisson-based models, other studies built more detailed ways to describe how the order book changes over time. Bacry, Mastromatteo, and Muzy (2015) used Hawkes processes to model how one trade or cancellation can make more orders appear soon after. This idea shows that market activity is not random. When trading becomes active, it often stays active for a while instead of just a second. The Hawkes model helps explain why orders, trades, and volatility come in clusters. It gives a more realistic picture of how markets move compared to simple Poisson models. Similarly, Cont and de Larrard (2013) studied the limit order book using a birth–death process. In their model, the bid and ask queues grow when new orders arrive. And they shrink when orders are traded or canceled. When one queue becomes empty, the price just moves. This way of modeling connects the change in price directly to how fast orders enter and leave the market. It also helps explain short-term price changes using simple queueing rules.

While the classical Hawkes process assumes that order arrivals only depend on past events, real financial markets are more complex. Traders react to current market conditions, and each order can carry different levels of information. To capture these behaviors, researchers have developed extended

versions of Hawkes models that make them more realistic for limit order book dynamics. One important extension is the state-dependent or queue-reactive Hawkes model, where the order arrival rate also depends on the current state of the book, such as the spread or order imbalance. Studies like Morariu-Patrichi and Pakkanen (2022) and Wu, Rambaldi, Muzy, and Bacry (2019) show that these models can capture short-term feedback between order flow and market conditions more accurately than the classical form. Another development is the marked or compound Hawkes model, which treats each event as carrying extra information like order size, price level, or trade direction. Rambaldi, Bacry, and Muzy (2017) and Swishchuk and Huffman (2018) found that this approach helps explain how different types of orders trigger more market activity.

As these stochastic and self-exciting models became more realistic, researchers also started to explore how individual traders should behave within such environments. Within these electronic stochastic books, optimal behaviour became an important question. Some researchers focused on how traders should act in these random market settings. Avellaneda and Stoikov (2008) built a model for market makers who must set bid and ask prices while managing their own inventory. Their method helps balance profit and risk by adjusting quotes when market activity or order flow changes. Obizhaeva and Wang (2013) studied how traders can split large orders into smaller ones to reduce cost. They made a resilient order book model, where the impact of a trade gets smaller as new orders refill the book. This model is similar to a queue that gets busy but then slowly clears up again.

Combined with the gradually matured agent-based and stochastic modelling research, Jericevich, Chang, and Gebbie (2021) implemented an asynchronous matching engine simulator called CoinTossX. Their goal is to reproduce realistic market dynamics, such as price formation, liquidity fluctuations, and volatility clustering, by allowing multiple types of trading agents to interact asynchronously in event time. Overall, the development of these models—from stochastic dynamics to optimal behavior and integrated simulations—reflects the continuous effort to bridge theory and real-world market mechanisms.

Section 2. Research Innovation

Our project plans to introduce the Hawkes self-exciting process into the existing CoinTossX agent-based simulation framework so that we could build a more realistic limit order book environment. Traditional models, such as Poisson-based systems, assume that order arrivals are independent. In contrast, our approach allows trading activities to cluster naturally through a Hawkes process mechanism, better capturing the bursty and dependent behavior seen in real markets. While previous work has used Hawkes processes mainly to generate synthetic order flows that are later matched by the engine, our approach goes one step further. **We plan to let the Hawkes intensity interact directly with agent behaviors, so that agents' trading decisions and market activity can influence each other dynamically.** This coupling between stochastic intensity and agent decision-making allows the model to better capture the feedback loops that often occur in real high-frequency markets, such as the clustering of orders, volatility bursts, and liquidity shifts.

Another innovation is that we plan to calibrate model parameters by using real tick data (for example, from the LOBSTER dataset). Parameters such as event arrival intensity, bid-ask spread distribution, order-book depth, and cancellation probabilities will be empirically estimated and integrated into the simulator. This data-driven calibration makes the simulation outputs closer to real market patterns and allows us to study how different levels of self-excitation affect liquidity, volatility, and price stability in a more interpretable way.

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